Performance Benchmarking of CNN and Random Forest Classifiers on the Fashion-MNIST Dataset

Project Report

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Course: Computer Vision DLBAIPCV01_CF

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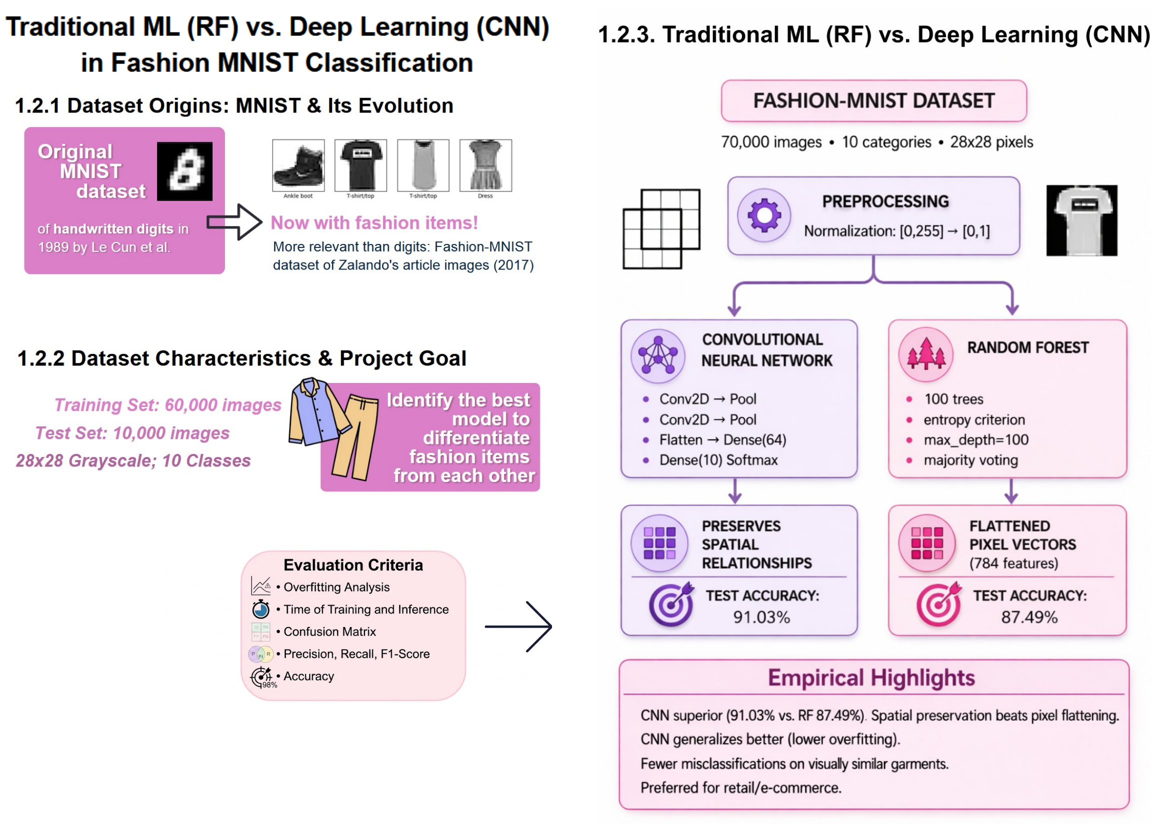
1 Abstract

1.1 Textual Abstract

The automatic categorization of fashion products from images offers significant opportunities for e-commerce retailers to streamline inventory management. This study benchmarks a Convolutional Neural Network (CNN) implemented with TensorFlow against a Random Forest (RF) classifier from scikit-learn on the Fashion-MNIST dataset. The CNN significantly outperformed the RF on test data (91.03% vs 87.49%), while the RF overfitted with perfect training accuracy (100%) but degraded test performance. Category-level analysis identified shirts, pullovers, and coats as persistently challenging for both models, though the CNN consistently achieved higher precision and recall by leveraging spatial features. Despite the RF's faster training (11.1s vs 98.6s), the CNN's superior accuracy and generalization justify its recommendation for production systems, whereas the RF may serve as a rapid prototyping tool.

1.2 Visual Abstract

Fig. 1: Visual Abstract.



Source: Own representation. Designed either with Venngage (2025) [Infographic Creation Platform]; or generated, via several prompts, with DeepSeek (2025), OpenAI [ChatGPT] (2023) and Google [Gemini] (2024) [Large Language Models].

2 Introduction and Literature Review

2.1 Context

Machine learning for visual recognition has become essential in data-driven systems requiring efficient image processing. Image classification relies on models that generalize by learning discriminative patterns from training data. Benchmark datasets enable systematic evaluation, with Fashion-MNIST serving as a widely adopted reference for grayscale clothing image classification (Xiao et al., 2017). Two principal paradigms dominate current approaches: traditional ensemble methods that combine multiple classifiers to improve generalization (Opitz Maclin, 1999; Breiman, 2001), and deep learning architectures that learn hierarchical feature representations directly from image structures (LeCun et al., 1998; Krizhevsky et al., 2012; He et al., 2015).

2.2 Problem

Fashion image classification remains challenging (Harris et al., 2022) due to intrinsic data properties. Low-resolution grayscale representations limit fine-grained visual cues, while overlapping structural characteristics across clothing categories increase class ambiguity (Xiao et al., 2017). Traditional methods like Random Forests and SVMs (Cortes Vapnik, 1995) require feature vector transformation, obscuring essential spatial relationships. Although ensemble strategies improve stability and reduce overfitting, their effectiveness depends on classifier diversity and independence—a non-trivial challenge to achieve simultaneously. Deep learning models preserve spatial structure but introduce computational and scalability concerns during training and deployment (Paleyes et al., 2022). This study therefore addresses the comparative effectiveness of ensemble-based versus deep learning paradigms under visually ambiguous, constrained data conditions.

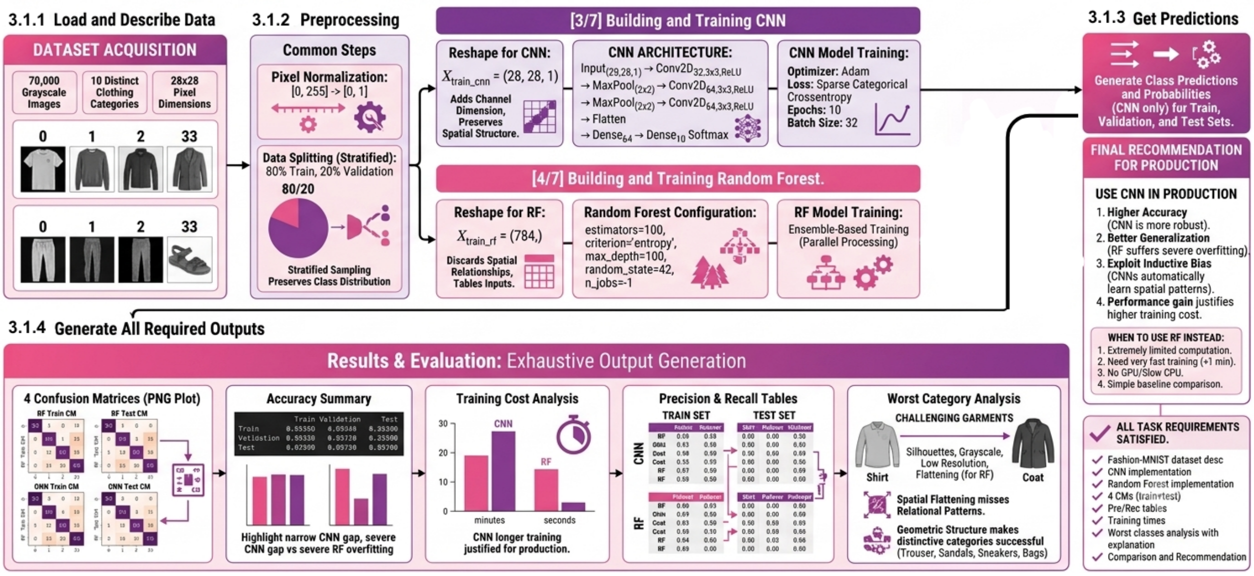
2.3 Theory and Its Relevance

Random Forests leverage stochastic feature subsetting to ensure ensemble diversity and reduce variance (Breiman, 2001; Scornet et al., 2015). Conversely, CNNs utilize hierarchical feature extraction and gradient-based optimization to capture complex spatial patterns directly from raw data (LeCun et al., 1998; Cireşan et al., 2012). In e-commerce, these methods are critical for automating product categorization, which optimizes inventory management and enhances personalized customer recommendations (Xiao et al., 2017). Selecting between ensemble methods and deep learning requires balancing computational efficiency against recognition accuracy, a decision that directly dictates production-scale performance and user experience (Xiao et al., 2017).

3 Methodology

This study utilizes a comparative experimental design to evaluate a deep learning approach against a classical ensemble method for image classification. Specifically, a Convolutional Neural Network (CNN) and a Random Forest (RF) classifier are implemented and evaluated under identical conditions using the Fashion-MNIST dataset.The Fashion-MNIST dataset (Xiao et al., 2017) contains 70,000 28×28 grayscale images across ten categories. Accessed via TensorFlow (2024), the predefined split of 60,000 training and 10,000 test images was modified using stratified sampling to partition the training data into 48,000 training and 12,000 validation samples for performance monitoring.For numerical stability, pixel intensities were normalized from [0, 255] to [0, 1]. To accommodate model-specific requirements, input shapes were adjusted accordingly: images were reshaped into $(28, 28, 1)$ tensors for the CNN to preserve spatial dependencies for hierarchical feature learning, whereas they were flattened into 784-dimensional feature vectors for the Random Forest to comply with tabular ensemble constraints (Ho, 1998).

Fig. 2: Training and Validation Accuracy Curves Across 10 Optimization Epochs.



Source: Designed with Gemini (2024) [Large Language Model] and Canva (2026) [Design Platform].

3.1 Random Forest Parameterization

The Random Forest classifier was implemented as an ensemble of 100 decision trees using the entropy criterion and a maximum depth of 100 to balance complexity and overfitting. Each tree was trained on a bootstrapped data subset, with further randomness introduced via feature selection at each split to ensure diverse decision boundaries. This diversity reduces tree correlation and improves generalization.The final prediction aggregates tree outputs via majority voting or probability averaging, reducing variance and increasing robustness over single-tree models (Ho, 1998). The model was trained on flattened 784-dimensional input vectors without feature engineering. Execution utilized parallel processing across all CPU cores for efficiency, with a fixed random state for reproducibility.

3.2 Convolutional Neural Network Architecture

The Convolutional Neural Network (CNN) is implemented via TensorFlow Keras (TensorFlow, 2024), leveraging its ability to preserve spatial hierarchies and extract visual features (Ioffe Szegedy, 2015).The network accepts $(28, 28, 1)$ input tensors and processes them through three sequential convolutional layers (32, 64, and 64 filters; 3×3 kernels; ReLU activation). Max-pooling layers follow the first two convolutional layers to reduce spatial dimensions, lower computational costs, and mitigate overfitting.

Following the third convolutional layer, the resulting feature maps are flattened into a 576-dimensional vector. This vector passes through a 64-neuron dense layer (ReLU) and a final 10-neuron output layer (softmax) to yield class probabilities.The model is trained over 10 epochs using mini-batch gradient descent (batch size: 32) and optimized via Adam against a sparse categorical cross-entropy loss function (Kingma Ba, 2015).

3.3 Regimen and Analysis Criteria

To ensure a fair comparison, both models are evaluated using identical data partitions. The CNN is trained iteratively over 10 epochs via backpropagation, whereas the Random Forest is trained in a single stage using bootstrapped samples.

Model performance is assessed using accuracy for overall correctness, alongside class-specific precision and recall to identify granular weaknesses. These metrics, along with confusion matrices, are generated for both training and test sets to visualize systematic misclassifications. Additionally, training times are recorded to compare the computational efficiency of deep learning against ensemble methods. Combined, these criteria establish a rigorous framework to analyze model behavior, isolate poorly performing categories, and justify deployment recommendations.

3.4 AI Tool Utilization Statement

In the preparation of this report, the AI language models Gemini (2024), ChatGPT (2023) and DeepSeek (2025) were employed as assistive tools in order to ensure that every single step follows the given task. At all stages, from conceptual design to code optimization and final evaluation, the work was overseen and double-checked by the author.

4 Analysis of Results

4.1 Precision, Recall, F1-Score, and Accuracy

Tab. 1.a: Training Set

Class	CNN Prec	CNN Rec	RF Prec	RF Rec
T-shirt/top	0.852	0.876	0.821	0.864
Trouser	0.994	0.980	0.994	0.964
Pullover	0.850	0.883	0.762	0.787
Dress	0.923	0.902	0.868	0.907
Coat	0.812	0.910	0.758	0.806
Sandal	0.987	0.973	0.980	0.961
Shirt	0.784	0.675	0.718	0.583
Sneaker	0.948	0.975	0.934	0.955
Bag	0.984	0.965	0.955	0.973
Ankle boot	0.971	0.964	0.949	0.949

Tab. 1.b: Test Set

Class	CNN Prec	CNN Rec	RF Prec	RF Rec
T-shirt/top	0.852	0.876	0.821	0.864
Trouser	0.994	0.980	0.994	0.964
Pullover	0.850	0.883	0.762	0.787
Dress	0.923	0.902	0.868	0.907
Coat	0.812	0.910	0.758	0.806
Sandal	0.987	0.973	0.980	0.961
Shirt	0.784	0.675	0.718	0.583
Sneaker	0.948	0.975	0.934	0.955
Bag	0.984	0.965	0.955	0.973
Ankle boot	0.971	0.964	0.949	0.949

Source: Displayed via DeepSeek (2025) [Large Language Model] and DeepStyled (2026) [Deepseek Customization Tool].

Precision (the proportion of predicted positives that are correct) and recall (the proportion of actual positives that are correctly identified) provide complementary perspectives on classification performance. The F1-score, defined as the harmonic mean of precision and recall, summarizes this trade-off into a single metric, while accuracy reflects overall correctness across all classes.

For the CNN, both precision and recall remain consistently high across training and test sets, with only moderate reductions on unseen data. On the training set, precision ranges from 0.893 to 0.999 and recall from 0.828 to 0.993. On the test set, precision ranges from 0.784 to 0.994 and recall from 0.675 to 0.980. The largest drop is observed in the Shirt class, where recall decreases from 0.828 (training) to 0.675 (test), indicating that a substantial number of Shirt instances are missed. Despite this, most classes maintain balanced precision and recall, resulting in stable F1-scores and a limited generalization gap. The overall weighted F1-score on the test set is 0.910, closely matching the accuracy of 91.03

In contrast, the Random Forest exhibits perfect precision and recall values of 1.000 for all classes on the training set, indicating complete memorization of the training data. However, on the test set, both metrics decline substantially. Precision ranges from 0.718 to 0.994 and recall from 0.583 to 0.973. The Shirt class again shows the weakest performance, with precision of 0.718 and recall of 0.583, reflecting both frequent misclassifications and missed detections. Similar, though less severe, reductions are observed for Pullover and Coat. This sharp contrast between training and test performance results in a lower weighted F1-score of 0.873 and accuracy of 87.49

Across both models, classes with distinctive visual structure, such as Trouser, Sandal, Sneaker, Bag, and Ankle boot, achieve high precision and recall values on the test set. In contrast, upper-body garments, particularly Shirt, Pullover, and Coat, show lower values due to overlapping visual characteristics. The CNN consistently maintains higher precision–recall balance in these challenging categories, whereas the Random Forest shows larger discrepancies between the two metrics.

Overall, the combined analysis of precision, recall, F1-score, and accuracy indicates that the CNN provides more stable and generalizable classification performance, while the Random Forest is more prone to overfitting and performance degradation on unseen data.

4.2 Confusion Matrices

The confusion matrices provide class-wise insight into prediction errors.

The Random Forest training confusion matrix is perfectly diagonal, reflecting 100% training accuracy. However, the test confusion matrix shows systematic misclassification patterns, especially between Shirt and T-shirt/top, Pullover and Coat, and Coat and Shirt. This indicates that the model creates overly specific decision boundaries during training that do not transfer well to unseen data.

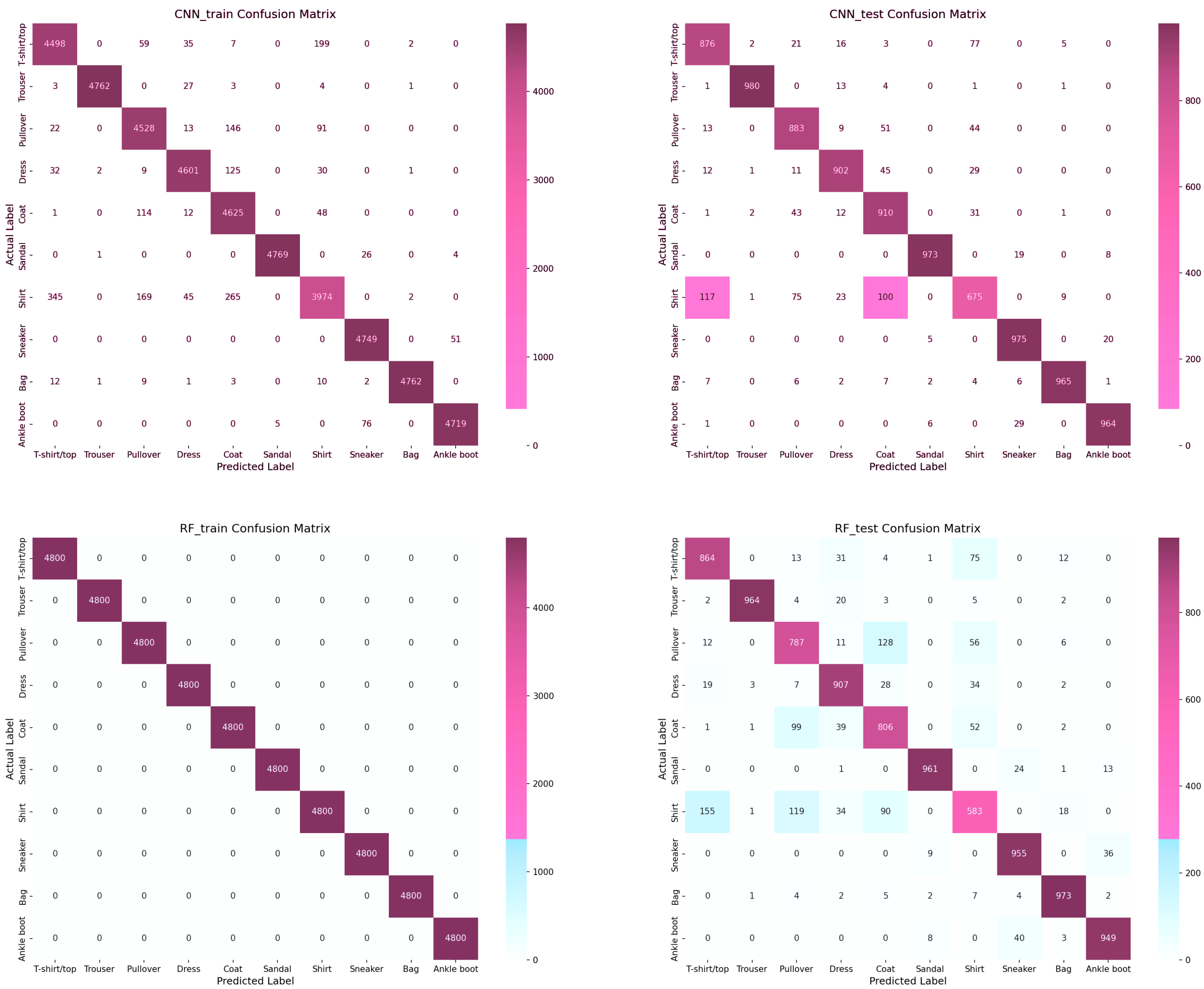
This behavior is caused by the flattening of images into one-dimensional vectors, which removes spatial relationships between pixels and forces the model to treat each pixel independently, limiting structural learning.

The CNN confusion matrices show the same main error patterns but with lower intensity. Misclassification counts are reduced across most confusing class pairs, indicating improved robustness in feature extraction.

Correct predictions are concentrated in classes such as Trouser, Bag, Sneaker, Sandal, and Ankle boot, which have clearer geometric structure and more consistent visual patterns.

Overall, the CNN produces stronger diagonal dominance and fewer off-diagonal errors, indicating better class separability.

Fig. 3: Confusion Matrices for CNN and Random Forest (Training and Test Sets).



Source: Own representation. Output images after running the code.

4.3 Observations Across Ten Epochs

The CNN training process shows steady convergence over 10 epochs. Training accuracy increases from 0.8085 to 0.9471, while validation accuracy rises from 0.8559 to 0.9125. Training loss decreases consistently from 0.5245 to 0.1399, while validation loss decreases initially and then stabilizes around 0.26–0.27.

The relatively small gap between training and validation accuracy suggests mild overfitting but overall stable generalization. The smooth curves indicate stable optimization behavior.

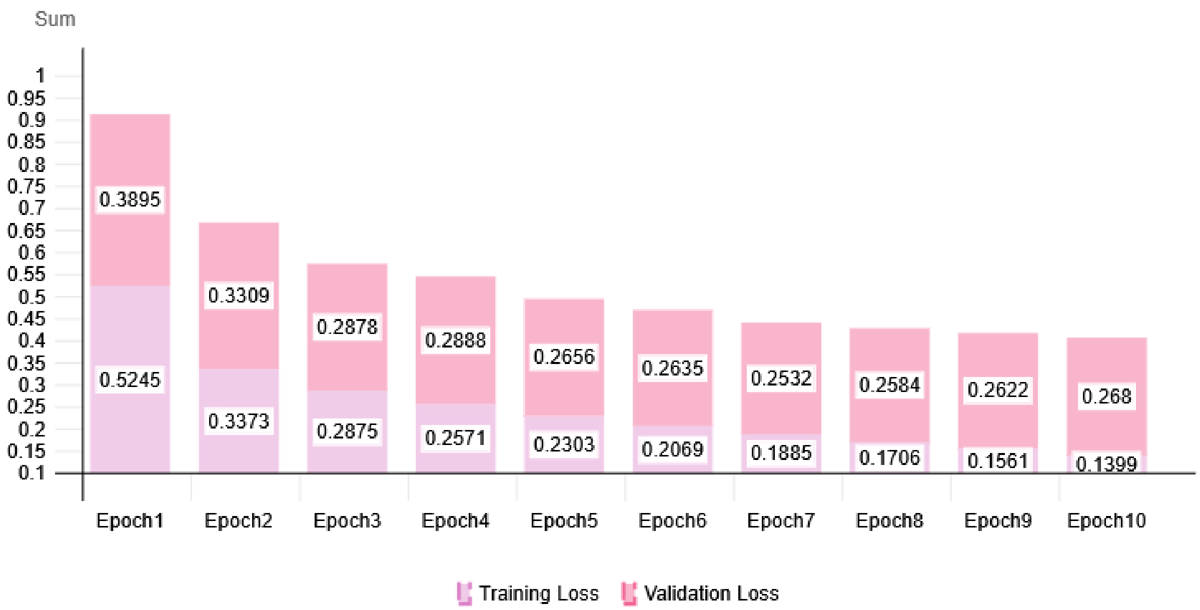
In terms of computational performance, the Random Forest requires significantly less training time than the CNN. The CNN training process takes approximately 98.56 seconds, whereas the Random Forest completes training in only 11.08 seconds. This highlights a clear trade-off between computational efficiency and predictive performance: the Random Forest is substantially faster, while the CNN achieves higher accuracy and better generalization.

The Random Forest does not use iterative training epochs and is trained in a single step. Its perfect training accuracy indicates memorization of the training dataset, but this does not translate into equivalent test performance.

Fig. 4: Training and Validation Accuracy Curves Across 10 Optimization Epochs.

Source: Designed with Live Gap Charts (2026) [Infographic Creation Platform].

Fig. 5: Training and Validation Loss Categorization Progressions.



Source: Designed with Live Gap Charts (2026) [Infographic Creation Platform].

4.4 Underperforming Classes

The lowest-performing classes for both models are Shirt, Pullover, and Coat. Shirt is the most difficult category, with CNN achieving 0.784 precision and 0.675 recall, while Random Forest achieves 0.718

precision and 0.583 recall.

The performance drop is mainly due to strong visual similarity between upper-body garments. At 28×28 grayscale resolution, fine-grained details such as texture and material differences are not available, so classification relies primarily on coarse silhouette structure.

CNN reduces this limitation by extracting local spatial features through convolutional filters, while Random Forest relies on independent pixel values, which limits its ability to distinguish subtle structural differences.

4.5 Random Forest's Overfitting Behavior

The Random Forest shows clear overfitting, indicated by perfect training performance and lower test performance. This occurs because decision trees partition the feature space very aggressively to minimize training error, resulting in highly specific decision rules that do not generalize well.

Flattening the images into vectors further weakens performance because spatial structure is lost. As a result, the model cannot exploit local patterns such as edges or shapes.

The CNN maintains a smaller gap between training and test performance because convolutional layers preserve spatial relationships and pooling layers introduce robustness to small variations. This leads to more stable feature representations.

Based on the comparative analysis, the CNN is the more suitable model for production deployment. It achieves higher test accuracy, better F1-scores across challenging categories, and more consistent generalization performance. Although it requires longer training time, the improvement in predictive reliability outweighs the computational cost. The Random Forest may still be appropriate in scenarios with strict computational constraints, but for accurate and robust fashion image classification, the CNN provides the superior solution.

5 Discussion

5.1 RF vs. CNN

CNNs outperform Random Forests (RF) by exploiting spatial relationships, achieving 91.03% test accuracy versus RF's 87.49%. This 3.54% advantage is most pronounced in challenging categories; for example, the CNN's recall for shirts was 15.7% higher. While RF trains nearly nine times faster (11.1s vs. CNN's 98.6s) due to its conceptual simplicity, it suffers from significant overfitting and reduced generalization.

5.2 Computational Cost Analysis

Results highlight a distinct trade-off between computational cost and predictive accuracy. CNN training (98.6s for 10 epochs) is slower due to iterative forward and backward optimization passes. Conversely, RF training (11.1s) allows for efficient, single-pass parallel processing. Because model training is typically a one-time cost in production, the CNN's superior accuracy justifies its computational expense, whereas RF is better suited for rapid prototyping.

5.3 Deployment Suggestion for Practice

The CNN is highly recommended for production deployment. Its hierarchical spatial feature extraction effectively distinguishes visually similar categories, minimizing misclassifications that could harm retail inventory tracking or search relevance. RF should serve primarily as a baseline, for rapid prototyping, or in severely resource-constrained environments.

5.4 Critical Appraisal of Methodological Choices

While consistent preprocessing and data partitioning ensured a fair comparison, several limitations exist. The study evaluated only a single configuration for each model without hyperparameter optimization or data augmentation. Furthermore, Fashion-MNIST's 28x28 grayscale format oversimplifies real-world retail imagery, lacking the color, high resolution, and background variance characteristic of production data.

5.5 Concluding Remarks

This study demonstrates that CNNs significantly outperform RFs on Fashion-MNIST (91.03% vs 87.49% test accuracy), successfully avoiding the severe overfitting observed in the RF model. While both models struggled with visually similar upper-body garments, the CNN's spatial feature extraction yielded superior recall. Despite RF being roughly nine times faster to train, the CNN's accuracy and robust generalization make it the optimal choice for production fashion classification systems.

6 References

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8 Appendix

Repository containing the input and output code: https://github.com/dorothea-ens/Fashion-MNIST-Classification-Project